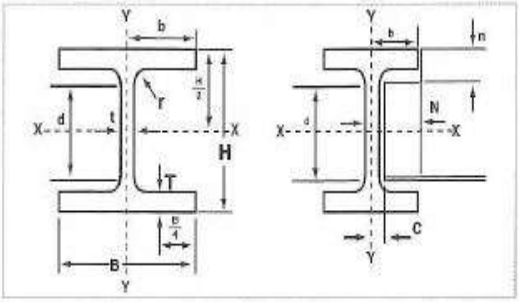
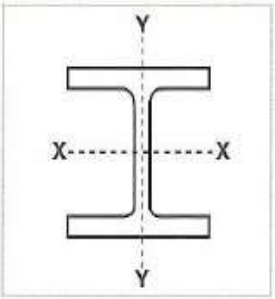


4a (i) Beams (Imperial Units) - Part I

	Designation Size		Unit Weight		Height Of Beam	Width Of Beam	Thickness		Root Radius	Depth Between Fillets	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
							Flange	Web				Flange	Web	End Clearance	Notch	
					H	B	T	t	r	d	A	b/T	d/t	C	N	n
	in	(mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W27	27 x 10	(686 x 254)	84	125.0	677.9	253.0	16.2	11.7	15.2	615.1	159	7.81	52.6	8	131	31
			94	139.9	683.5	253.7	19.0	12.4	15.2	615.1	178	6.68	49.6	8	131	34
			102	151.8	687.5	254.5	21.0	13.2	15.2	615.1	194	6.06	46.6	9	131	36
			114	169.7	692.9	255.8	23.7	14.5	15.2	615.1	217	5.40	42.4	9	131	39
			129	192.0	701.8	254.3	27.9	15.5	15.2	615.6	244	4.56	39.7	10	129	43
			143	212.8	707.9	255.8	31.0	17.0	15.2	615.5	270	4.13	36.2	11	129	46
			159	236.6	714.0	257.8	34.0	19.1	15.2	615.6	301	3.79	32.2	12	129	49
			182	270.8	723.9	260.2	39.1	21.6	15.2	615.3	345	3.33	28.5	13	129	54
			201	299.1	732.0	262.8	42.9	23.9	15.2	615.8	382	3.06	25.8	14	129	58
			221	328.9	739.9	264.8	47.0	25.9	15.2	615.5	418	2.82	23.8	15	129	62
			247	367.6	749.8	267.7	52.1	28.9	15.2	615.2	468	2.57	21.3	16	129	67
			271	403.3	760.0	270.3	56.9	31.5	15.2	615.8	513	2.38	19.5	18	129	72
			302	449.4	771.9	273.8	63.0	35.1	15.2	615.5	574	2.17	17.5	20	129	78
	27 x 14	(686 x 356)	146	217.3	695.5	354.7	24.8	15.4	15.2	615.6	277	7.16	40.1	10	180	40
			161	239.6	700.8	356.1	27.4	16.8	15.2	615.5	306	6.49	36.7	10	180	43
			178	264.9	706.4	357.8	30.2	18.4	15.2	615.5	337	5.92	33.4	11	180	45
			194	288.7	714.0	356.4	34.0	19.1	15.2	615.5	368	5.24	32.3	12	179	49
			217	322.9	722.1	358.5	38.1	21.1	15.2	615.5	411	4.70	29.2	13	179	53
			235	349.7	728.0	360.4	40.9	23.1	15.2	615.8	446	4.41	26.6	14	179	56
			258	383.9	736.1	362.5	45.0	24.9	15.2	615.8	489	4.03	24.7	14	179	60
			281	418.2	744.0	364.5	49.0	26.9	15.2	615.6	533	3.72	22.9	15	179	64
			307	456.9	752.1	366.9	53.1	29.5	15.2	615.5	582	3.46	20.9	17	179	68
			336	500.0	762.0	369.0	57.9	32.0	15.2	615.8	636	3.19	19.2	18	179	73
W30	30 x 10 1/2	(762 x 267)	90	133.9	750.0	264.4	15.5	12.0	16.5	686.0	171	8.53	57.2	8	136	32
			99	147.3	754.0	265.2	17.5	12.8	16.5	686.0	187	7.58	53.6	8	136	34
			108	160.7	757.7	266.1	19.3	13.8	16.5	686.1	205	6.89	49.6	9	136	36
			116	172.6	762.2	266.7	21.6	14.3	16.5	686.0	220	6.17	48.0	9	136	38
			124	184.5	766.3	267.1	23.6	14.9	16.5	686.1	235	5.65	46.2	9	136	40
			132	196.4	769.8	268.0	25.4	15.6	16.5	686.0	251	5.28	44.0	10	136	42
			148	220.2	779.0	266.2	30.0	16.5	16.5	686.0	281	4.44	41.6	10	135	47
			165	245.5	785.1	268.1	33.0	18.5	16.5	686.1	312	4.06	37.1	11	135	50
			185	275.3	793.0	270.1	37.1	20.6	16.5	685.8	351	3.64	33.3	12	135	54
			207	308.0	801.1	272.7	40.9	23.1	16.5	686.3	392	3.33	29.7	14	135	57
			226	336.3	809.0	274.6	44.9	24.9	16.5	686.2	428	3.06	27.6	14	135	61
			246	366.1	817.1	276.6	48.0	26.9	16.5	688.1	462	2.88	25.6	15	135	65
			269	400.3	825.0	279.1	53.1	29.5	16.5	685.8	511	2.63	23.2	17	135	70
			295	439.0	835.2	281.7	57.9	32.0	16.5	686.4	559	2.43	21.5	18	135	74
	30 x 15	(762 x 381)	173	257.5	773.2	380.6	27.1	16.6	16.5	686.1	328	7.04	41.2	10	192	44
			191	284.2	779.3	382.0	30.1	18.0	16.5	686.1	362	6.35	38.1	11	192	47
			211	314.0	785.9	383.7	33.4	19.7	16.5	686.1	400	5.74	34.9	12	192	50
			235	349.7	795.0	382.4	38.1	21.1	16.5	685.8	445	5.02	32.5	13	191	55
			261	388.4	802.9	384.9	41.9	23.6	16.5	686.1	495	4.59	29.0	14	191	58
			292	434.5	813.1	387.5	47.0	25.9	16.5	686.1	553	4.12	26.5	15	191	63
			326	485.1	823.0	390.4	52.1	29.0	16.5	685.9	617	3.75	23.7	16	191	69
			357	531.3	833.0	393.0	56.9	31.5	20.0	679.2	677	3.45	21.6	18	191	77

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
*Please contact us for further clarification.

Dimensions and Properties



4a (i) Beams (Imperial Units) - Part II

Unit Weight		Surface Area per Metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation Size	
			Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J		
lb/ft	kg/m	m^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3			dm^6	cm^4	in	
84	125.01	2.32	118000	4383	27.2	5.24	3481	347	3994	542	0.862	43.9	4.80	116	27 x 10	W27
94	139.89	2.33	136300	5184	27.6	5.39	3987	409	4558	638	0.868	38.7	5.72	169		
102	151.79	2.34	150400	5784	27.8	5.46	4374	455	5000	710	0.871	35.5	6.42	220		
114	169.65	2.35	170300	6631	28.0	5.53	4916	518	5631	811	0.872	31.8	7.42	308		
129	191.97	2.36	198000	7670	28.5	5.61	5644	603	6462	943	0.878	27.7	8.71	462		
143	212.81	2.38	222000	8678	28.7	5.67	6272	678	7204	1063	0.878	25.2	9.94	627		
159	236.62	2.39	247800	9750	28.7	5.69	6940	756	8016	1191	0.877	23.0	11.3	840		
182	270.85	2.42	289300	11540	29.0	5.78	7993	887	9282	1402	0.878	20.3	13.5	1260		
201	299.12	2.44	323800	13060	29.1	5.85	8847	994	10330	1577	0.877	18.6	15.5	1675		
221	328.88	2.46	359400	14640	29.3	5.92	9715	1106	11390	1759	0.877	17.2	17.6	2186		
247	367.58	2.49	406900	16800	29.5	5.99	10850	1255	12810	2005	0.877	15.6	20.4	2993		
271	403.29	2.51	453800	18910	29.7	6.07	11940	1399	14170	2243	0.877	14.4	23.4	3897		
302	449.42	2.54	515400	21790	30.0	6.16	13350	1592	15950	2565	0.876	13.2	27.4	5322		
146	217.27	2.75	234200	18440	29.1	8.16	6735	1040	7558	1598	0.884	29.7	20.7	454	27 x 14	
161	239.59	2.77	261200	20670	29.2	8.22	7455	1161	8389	1787	0.885	27.0	23.4	610		
178	264.89	2.78	290800	23120	29.4	8.28	8233	1292	9297	1992	0.885	24.6	26.4	813		
194	288.70	2.79	325500	25720	29.8	8.36	9118	1444	10300	2223	0.889	22.3	29.7	1100		
217	322.93	2.81	369200	29310	30.0	8.44	10230	1635	11610	2523	0.890	20.1	34.3	1532		
235	349.72	2.83	402300	31970	30.0	8.47	11050	1774	12600	2745	0.888	18.8	37.7	1914		
258	383.95	2.85	447800	35780	30.3	8.56	12170	1974	13930	3057	0.889	17.2	42.7	2518		
281	418.17	2.87	494700	39680	30.5	8.63	13300	2177	15290	3377	0.890	15.9	47.9	3250		
307	456.87	2.89	545000	43850	30.6	8.68	14990	2390	16750	3717	0.889	14.8	53.6	4152		
336	500.02	2.91	604800	48670	30.8	8.75	15870	2638	18450	4111	0.889	13.7	60.3	5373		
90	133.93	2.51	150700	4788	29.7	5.30	4018	362	4644	570	0.854	49.8	6.46	119	30 x 10 1/2	W30
99	147.33	2.51	168500	5456	30.0	5.40	4470	411	5156	647	0.858	45.2	7.40	159		
108	160.72	2.52	185900	6080	30.1	5.45	4907	457	5665	720	0.859	41.5	8.29	208		
116	172.63	2.53	205300	6850	30.5	5.58	5387	514	6198	807	0.864	38.1	9.39	267		
124	184.53	2.54	223100	7525	30.8	5.65	5821	563	6689	885	0.867	35.4	10.4	333		
132	196.44	2.55	240000	8175	30.9	5.71	6234	610	7167	959	0.869	33.2	11.3	404		
148	220.25	2.56	278200	9463	31.5	5.81	7142	711	8197	1115	0.875	28.9	13.3	605		
165	245.55	2.58	310700	10640	31.5	5.84	7914	794	9129	1251	0.874	26.4	15.0	815		
185	275.31	2.60	353200	12240	31.7	5.91	8909	906	10320	1433	0.874	23.7	17.5	1147		
207	308.05	2.62	397200	13900	31.8	5.96	9916	1020	11550	1620	0.873	21.6	20.1	1555		
226	336.32	2.64	440500	15600	32.1	6.04	10890	1136	12720	1808	0.874	19.9	22.8	2029		
246	366.09	2.66	480200	17050	32.2	6.08	11750	1233	13790	1971	0.873	18.7	25.2	2500		
269	400.32	2.68	536500	19400	32.4	6.16	13010	1390	15330	2229	0.874	17.0	28.9	3361		
295	439.01	2.70	595900	21780	32.7	6.24	14270	1546	16900	2486	0.874	15.8	32.9	4345		
173	257.45	3.01	341200	24890	32.3	8.71	8827	1308	9916	2012	0.883	30.3	34.6	635	30 x 15	
191	284.24	3.02	381700	28000	32.5	8.80	9796	1466	11030	2258	0.885	27.5	39.3	859		
211	314.00	3.04	427000	31500	32.7	8.87	10870	1642	12270	2531	0.886	25.0	44.6	1161		
235	349.72	3.05	485900	35570	33.0	8.94	12220	1860	13830	2869	0.889	22.2	50.9	1651		
261	388.41	3.07	543700	39920	33.1	8.98	13540	2074	15410	3208	0.888	20.3	57.8	2218		
292	434.54	3.10	618300	45680	33.4	9.09	15210	2358	17380	3653	0.889	18.3	67.0	3090		
326	485.14	3.12	697600	51790	33.6	9.16	16950	2653	19500	4123	0.888	16.7	77.0	4224		
357	531.27	3.14	776600	57770	33.9	9.24	18650	2940	21550	4579	0.888	15.3	87.0	5563		

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
*Please contact us for further clarification.